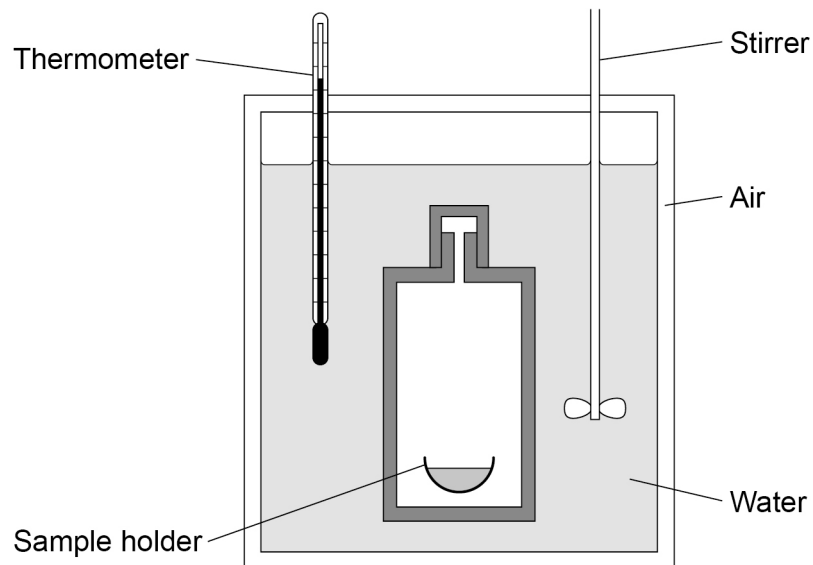


Figure 2 shows one type of calorimeter.

Figure 2



A calorimeter can be used to determine the chemical energy store of biomass. A known mass of biomass is fully combusted in a calorimeter. The heat energy released from this combustion increases the temperature of the water in the calorimeter. The increase in the temperature of a known volume of water is recorded.

03.1

Other than the thermometer, explain how **two** features of the calorimeter shown in **Figure 2** would enable a valid measurement of the total heat energy released.

[2 marks]

- 1 _____
- _____
- _____
- 2 _____
- _____
- _____

Question 3 continues on the next page

0	3	.	2
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A 2 g sample of biomass was fully combusted in a calorimeter.

The volume of water in the calorimeter was 100 cm³

The increase in temperature recorded was 15.7 °C

4.18 J of energy are needed to increase the temperature of 1 cm³ of water by 1 °C

Use this information to calculate the heat energy released in kJ per g of biomass.

Show your working.

[2 marks]

Answer _____ kJg⁻¹

Plants and algae produce fuels called biofuels. Scientists have used *Chlorella* to produce biofuel. *Chlorella* is a genus of single-celled photosynthetic alga. *Chlorella* can be grown in open ponds and fermenters.

0	3	.	3
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In natural ecosystems, most of the light falling on producers is **not** used in photosynthesis.

Suggest **two** reasons why.

[2 marks]

1 _____

2 _____

0 3 . 4 The light absorbed by chlorophyll is used in the light-dependent reaction.

Name the **two** products of the light-dependent reaction that are required for the light-independent reaction.

[2 marks]

1 _____

2 _____

0 3 . 5 *Chlorella* cells can divide rapidly. A culture of 2000 *Chlorella* cells was set up in a fermenter. The cells divided every 90 minutes.

You can assume that there were no limiting factors and that no cells died during the 24 hours.

Calculate the number of cells in the culture after 24 hours.

Give your answer in standard form.

Show your working.

[2 marks]

Answer _____

Turn over for the next question

